

Userstories

Customer (Online-Shop)

User Story	Involved Entities	Data Logic
Browse, filter, and search products	PRODUCT, CATEGORY	SELECT from Product where category_id matches or name LIKE search term.
View product details	PRODUCT, INVENTORY	Fetch product data and check Inventory for "In Stock" status.
Add/Remove from cart	ORDER_ITEM (Draft)	Temporary storage of product_id and quantity.
Registration / Login	CUSTOMER	INSERT or SELECT from Customer table (Auth).
View order history	ORDER, ORDER_ITEM, PRODUCT	JOIN Order with Order_Items and Products filtered by customer_id.

Backoffice/ ERP

User Story	Involved Entities	Data Logic
Warehouse inventory overview	WAREHOUSE, INVENTORY, PRODUCT, MATERIAL	Aggregate stock_level grouped by warehouse_id and item type.
Monitor customer orders	ORDER, ORDER_ITEM, CUSTOMER	List Orders with customer details and total quantity from Order_Items.
Track production cycles	PRODUCTION, PRODUCT, BILL_OF_MATERIALS	Monitor Production status; link to BOM to see required materials.
Order new goods (Purchase)	PURCHASE_ORDER, PURCHASE_ORDER_ITEM, SUPPLIER	INSERT into Purchase_Order and link items to Material.
Transfer goods between warehouses	INVENTORY, WAREHOUSE	Transaction: UPDATE Inventory (Decrement Whse A, Increment Whse B).

Analytics

User Story	Involved Entities	Data Logic
Live Sales Dashboard	ORDER, ORDER_ITEM	Real-time COUNT of Orders and SUM of sale_price for today.
Inventory Analytics	INVENTORY, MATERIAL, PRODUCT	Identify items where stock_level < threshold (Reorder Point).
Purchase Analysis	PURCHASE_ORDER_ITEM, SUPPLIER	Compare unit_cost over time per supplier and material.
Product Performance	ORDER_ITEM, PRODUCT, CATEGORY	GROUP BY product_id to find "Top 5 Best Sellers" by volume.